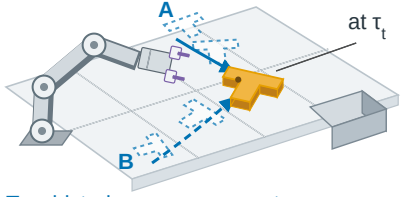
## Acting in physical time

Conceptual illustration; all positions and trajectories are hypothetical

### (a) Same pose, different motion



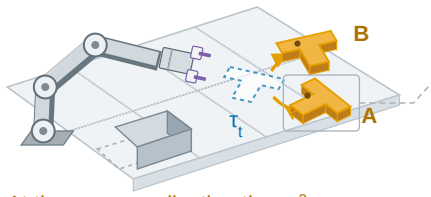

Two histories → one current pose

Past poses at −100, −50, 0 ms

—— A  ---- B

Current pose alone is ambiguous.

### (b) Different interaction poses



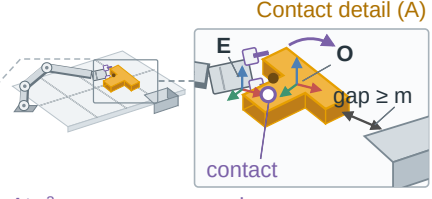

At the same application time $\tau^a$

A / B → different position + rotation

Predict its SE(3) pose at that time.

Carry the motion history forward.

### (c) Resolve contact geometry




At $\tau^a$: pose error ≤ ε and gap ≥ m

Target: tolerance ε, exact SE(3).

Context: margin m ≫ ε, 3D flow.

O: target; E: end-effector frame